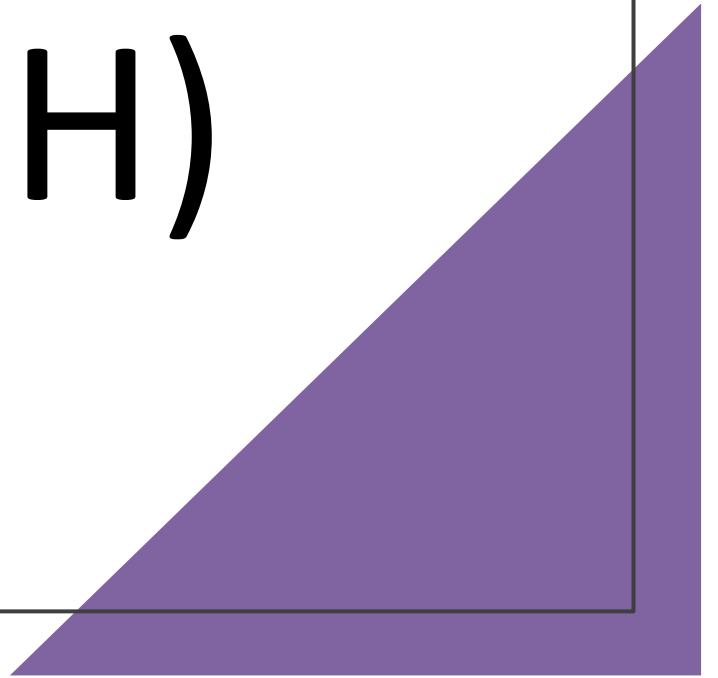


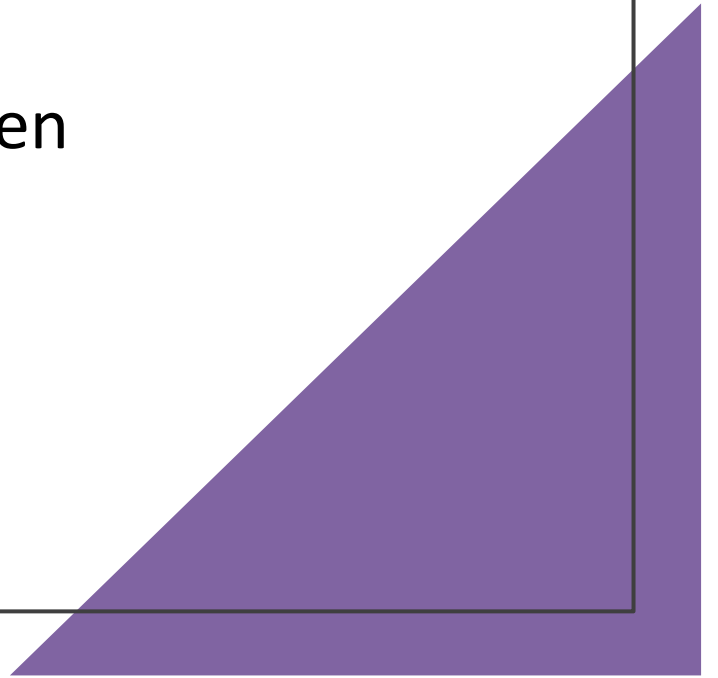
Benign Prostatic Hyperplasia (BPH)

Updated with Recent Urology Guidelines



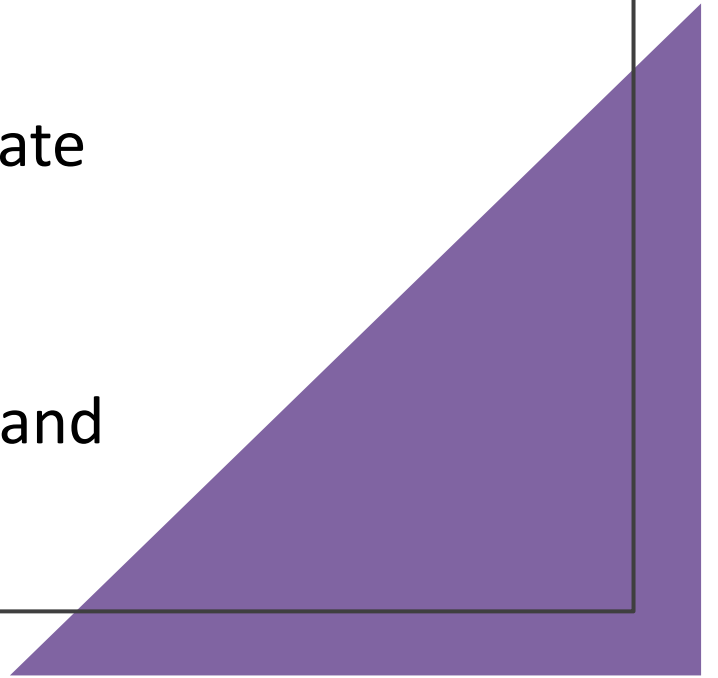
Definition & Epidemiology

- Benign Prostatic Hyperplasia (BPH): non-malignant enlargement of the prostate gland.
- Prevalence increases with age; affects ~50% of men over age 50 and 80% over age 80.
- Causes lower urinary tract symptoms (LUTS): obstructive and irritative symptoms.

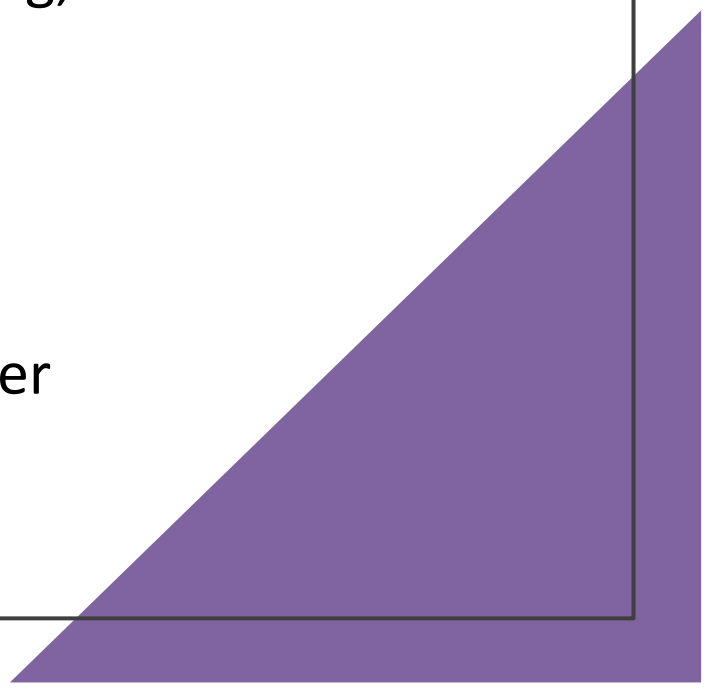


Pathophysiology

- Prostate enlargement due to stromal and epithelial cell proliferation.
- DHT (dihydrotestosterone) plays a major role in prostate growth.
- Bladder outlet obstruction leads to urinary retention and LUTS.



Clinical Presentation

- **Obstructive symptoms:** weak stream, hesitancy, straining, incomplete emptying.
 - **Irritative symptoms:** frequency, urgency, nocturia.
 - **Complications:** urinary retention, recurrent UTIs, bladder stones.
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Diagnostic Evaluation

- History and physical examination including digital rectal exam (DRE).
- International Prostate Symptom Score (IPSS).
- Urinalysis to rule out infection.
- Prostate-Specific Antigen (PSA) testing when indicated.
- Post-void residual (PVR) measurement and uroflowmetry.



International Prostate Symptom Score (IPSS)

The 7 IPSS Questions (Symptoms)

Symptom	Question
1. Incomplete emptying	Feeling bladder is not empty after urinating
2. Frequency	Urinating again <2 hours after last void
3. Intermittency	Flow stopping and starting
4. Urgency	Difficulty postponing urination
5. Weak stream	Reduced force of urinary stream
6. Straining	Need to push or strain to start
7. Nocturia	Number of times you get up at night to urinate

Each question is scored **0–5**, based on frequency in the past month:

Scoring for each question:

- **0** = Not at all
- **1** = Less than 1 time in 5
- **2** = Less than half the time
- **3** = About half the time
- **4** = More than half the time
- **5** = Almost always

Total Score	Severity
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0–7	Mild
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8–19	Moderate
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20–35	Severe
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Management Overview

- Treatment based on symptom severity and prostate size.
- Watchful waiting for mild symptoms.
- Pharmacologic therapy as first-line for moderate to severe symptoms.

TREATMENT

DESIRED OUTCOME

Reducing or eliminating obstructive and irritative voiding symptoms (primary goal).

Slowing disease progression

Preventing disease complications and reducing the need for surgical intervention.

Maintaining or improving quality of life.

General approach



Management options for BPH include

**Watchful waiting,
Drug therapy, and
Surgical intervention.**

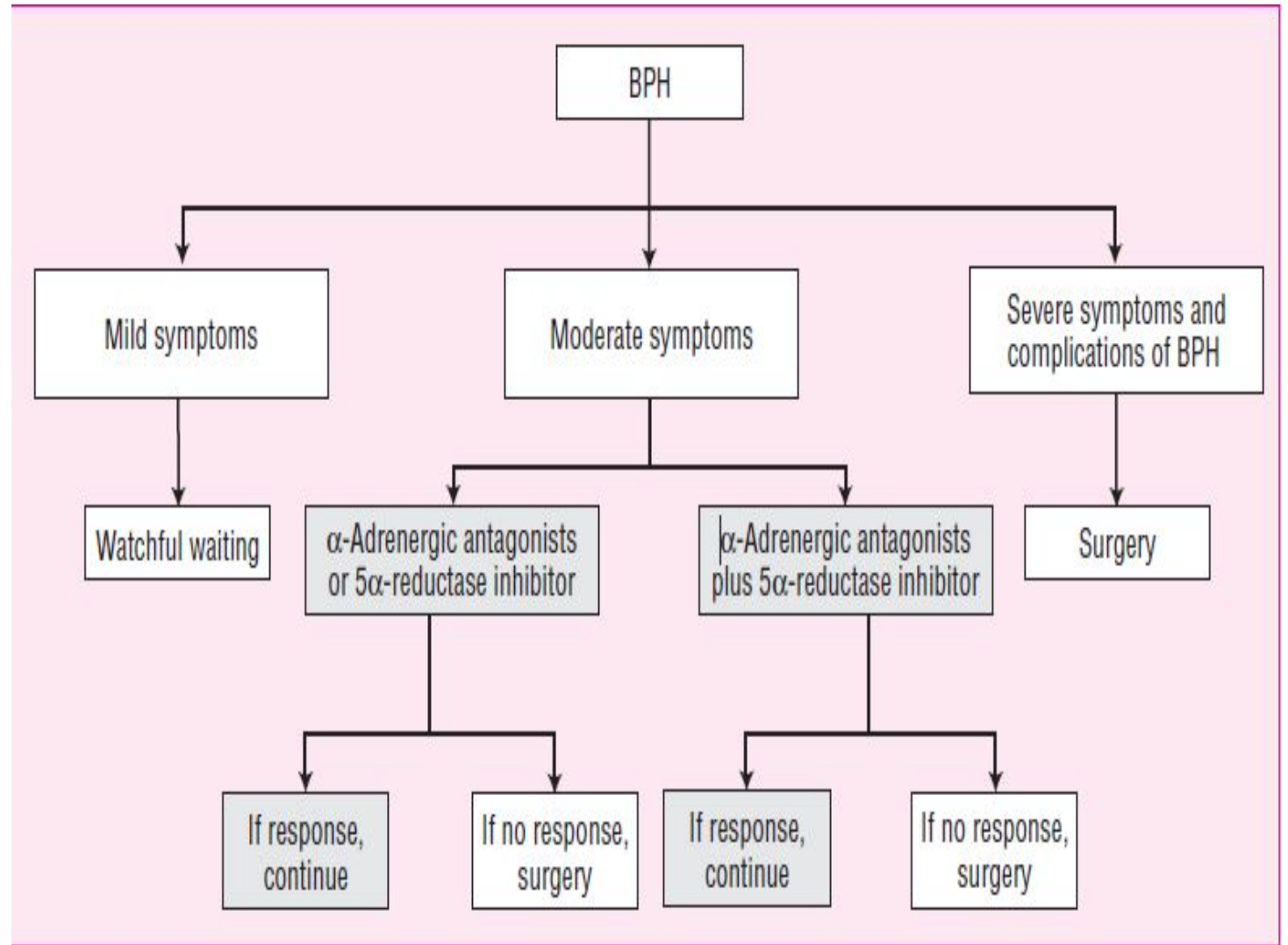


**The choice depends on the severity of
signs and symptoms.**



**Watchful waiting is appropriate for
patients with mild symptoms and for
those with moderate symptoms with
only mildly bothersome symptoms
and without complications.**

Algorithm for selection of treatment of BPH based on symptom severity



Pharmacologic
Treatment (Based
on
Pharmacotherapy
Handbook)

Alpha-blockers: tamsulosin, alfuzosin, doxazosin.

5-alpha-reductase inhibitors:
finasteride, dutasteride.

Combination therapy recommended
for large prostates (>40 mL).

Adverse effects: hypotension
(alpha-blockers), sexual dysfunction
(5-ARIs).

Updated Urology Recommendations (AUA/EAU 2023–2024)

- Alpha-blockers remain first-line for moderate to severe LUTS.
- 5 α -Reductase Inhibitors (5-ARIs) recommended for men with prostate enlargement or elevated PSA.
- Tadalafil 5 mg daily effective for LUTS, especially with ED.
- Minimally invasive procedures (UroLift, Rezūm) for selected patients.
- Surgery (Transurethral Resection of the Prostate=TURP) reserved for severe symptoms or complications.

α -Adrenergic Antagonists

- α -adrenergic antagonists relax the smooth muscle in the prostate and bladder neck, thereby increasing **urinary flow rates**
- **Tamsulosin** and **doxazosin** produce durable responses for 6 and 10 yrs, respectively.
- α -adrenergic antagonists do not decrease prostate volume or PSA levels.

α -Adrenergic Antagonists...

- 2nd generation (Terazosin, doxazosin, and **alfuzosin**) antagonize peripheral vascular α 1-adrenergic receptors in addition to those in the prostate.
 - Therefore, their adverse effects include **first dose syncope, orthostatic hypotension, and dizziness.**
- Alfuzosin is less likely to cause CV adverse effects than other second-generation agents.

α -Adrenergic Antagonists...

- Pts should be **slowly titrated to a maintenance dose** and should take these drugs at **bedtime** to minimize orthostatic hypotension and first-dose syncope with terazosin and doxazosin.
- Tamsulosin, the only 3rd-generation α -adrenergic antagonist, is selective for prostatic **α 1A-receptors**.
- Therefore, tamsulosin does **not cause peripheral vascular smooth muscle relaxation**.

α -Adrenergic Antagonists...

- Tamsulosin is a good choice for pts who
 - Cannot tolerate hypotension;
 - Have severe coronary artery disease, volume depletion, cardiac arrhythmias, severe orthostasis, or liver failure; or
 - Are taking multiple antihypertensives.
 - Want to avoid the delay of dose titration.

α -Adrenergic Antagonists...

- Caution is needed to avoid potential drug interactions.
 - Tamsulosin decreases metabolism of **cimetidine** and **diltiazem**.
 - **Carbamazepine** and **phenytoin** increase catabolism of α -adrenergic antagonists.

Testosterone → 5- α reductase → Dihydrotestosterone (DHT)

5 α -Reductase Inhibitors

- 5 α -Reductase inhibitors (**Dutasteride and Finasteride**) interfere with the stimulatory effect of testosterone.
- These agents slow disease progression and decrease the risk of complications.
- They do so by inhibiting 5 α -reductase which is responsible for intraprostatic conversion of testosterone to DHT, the active androgen which stimulates prostate tissue growth

5 α -Reductase Inhibitors....

- Compared with α -adrenergic antagonists, 5 α -reductase inhibitors have the disadvantages of
 - **Requiring 6 mths** to maximally shrink an enlarged prostate,
 - Being less likely to induce objective improvement, and
 - Causing more sexual dysfunction.

5 α -Reductase Inhibitors...

- **Dutasteride** inhibits types I & II 5 α -reductase, whereas finasteride inhibits only type II
- Dutasteride more quickly and completely suppresses intraprostatic DHT and decreases serum DHT by 90% (vs. 70%).
- 5 α -Reductase inhibitors may be preferred in pts with uncontrolled arrhythmias, poorly controlled angina, use of multiple antihypertensives, or inability to tolerate hypotensive effects of α -adrenergic antagonists.

5 α -Reductase Inhibitors...

- 5 α -Reductase inhibitors reduce serum PSA levels by 50%.
- PSA should be measured at baseline and repeated after 6 months.
- If **PSA** does not decrease by 50% after 6 mths of therapy in a compliant pt, the pt should be evaluated for prostate cancer.
- Both are FDA pregnancy category X.
 - Pregnant and potentially pregnant women should not handle the tablets or have contact with semen from men receiving 5 α -reductase inhibitors.

Surgical & Minimally Invasive Therapies

- TURP: Gold standard surgical treatment.
- Laser Therapies: HoLEP and GreenLight photovaporization.
- Prostatic Urethral Lift (PUL): preserves ejaculation.
- Water Vapor Therapy (Rezūm): minimally invasive outpatient option.

EVALUATION OF THERAPEUTIC OUTCOMES

- The primary therapeutic outcome of BPH therapy is
 - restoring adequate urinary flow without causing adverse effects.
- Outcome depends on
 - the pt's perception of effectiveness and
 - acceptability of therapy.

EVALUATION OF THERAPEUTIC OUTCOMES...

- **Objective measures** of bladder emptying (e.g., **uroflowmeter** and **postvoid residual urine** volumes) are also useful after 6 to 12 mths of 5 α -reductase inhibitor therapy or 3 to 4 wks of α -adrenergic antagonist therapy.
- **Laboratory tests** (e.g., blood urea nitrogen, creatinine, PSA) and urinalysis should be monitored regularly.
- In addition, pts should have an annual Digital Rectal Examination.

References

- Pharmacotherapy Handbook, 7th Edition.
- American Urological Association (AUA) BPH Guidelines, 2023.
- European Association of Urology (EAU) Guidelines on Male LUTS/BPH, 2024.